AI Evals for Engineers & PMs

Course syllabus · Instructor: Ion Moșnoi

A hands-on course on evaluating and improving AI products. Build a working agent, capture its behavior in traces, find its failures with a domain expert, and use evaluations to improve accuracy, cost and reliability.

Four weeks, with four live Saturday lectures, 10:00–12:00 Europe/Chișinău (UTC+2), plus 10+ hours of optional live office hours. Teaching is in English. Each module combines a lecture with a practical lab.

Module 1 — Building agents (Week 1)

Create the agent and traces that the rest of the course will use.

L1: AI fundamentals and the agent loop. Understand tokens, context limits, model behavior and structured outputs. Build a loop with tool calls, stop conditions and bounded retries.

Scope and permissions. Define allowed actions, when to ask for clarification and when to escalate to a human. Enforce permissions in code.

Make behavior visible. Record messages, model calls, tool inputs and outputs. Learn why a plausible final answer can hide a failed action.

Lab. Build a working agent and inspect its first complete trace.

Module 2 — Error analysis (Week 2)

Turn real failures into an error taxonomy and clear evaluation rules.

L2: Review with a domain expert. Build a trace viewer. Review varied examples, label Good or Bad, and note the exact step, evidence and expected behavior.

Find patterns. Cluster review notes into error types. Check category counts and prioritize failures by frequency and impact; an overall average can hide a broken feature.

One evaluator per failure mode. Write pass/fail criteria. Use code for objective checks and an LLM judge only when interpretation is needed.

Lab. Review traces, create an error taxonomy and write the rules your evals will check.

Module 3 — Evaluation data & judges (Week 3)

Build a useful test set and check whether the judge can be trusted.

L3: Create synthetic evaluation data. Combine features, roles and scenarios. Include ambiguous requests, missing information and multi-step tasks. Review questions, expected behavior and source evidence with an expert.

Separate development from testing. Version datasets, check category coverage and keep train, dev and test cases separate. Hold test cases aside while tuning prompts.

Calibrate the judge. Write judge prompts with clear rules and diverse examples. Compare judgments with expert labels, inspect false positives and false negatives, and report precision and recall by error type.

Lab. Create a versioned dataset and improve a judge using its disagreements with the expert.

Module 4 — Improve your agent (Week 4)

Use the failures to choose a fix, then measure what changed.

L4: Diagnose and experiment. Decide whether the problem is in data, retrieval, prompts, the model or tool flow. Try clearer instructions, better examples, query rewriting, routing or smaller task scopes.

Measure fixes and regressions. Compare versions on the same cases. Track failures by category alongside accuracy, cost and latency. Keep changes supported by the evidence.

Keep checking in production. Test prompt injection, permissions and unsafe actions. Automate regression checks in CI and monitor new failures and drift.

Lab. Run a before/after experiment and explain what improved, what regressed and what to try next.

How the course works

Who it is for. Engineers and PMs building AI products. Basic Python is useful for engineers; PMs can pair for coding labs. Use the supplied project or authorized, anonymized company data.

Practice and support. Plan 3–5 hours a week for lectures and practice, plus optional office hours arranged with the cohort. Includes six assignments with solutions and walkthroughs, a 200+ page reader, lifetime recordings and materials, Discord access and future live cohorts. Complete four assignments, including the final report, for the completion certificate.

Cohorts and fees. 21 November–18 December 2026: €500. 19 December 2026–15 January 2027: €1,000. All prices are in EUR. Teams of 2–20 save 10%.

Your instructor. Ion Moșnoi has 11+ years of experience in AI/ML, helping teams from startups to enterprises improve AI products. He also delivers personalized two-day workshops for engineering teams.